# DCP UNIVERSITY ENGINEERING CHALLENGE

**Engineering Solutions for Sustainable, Reliable and Efficient Cement Manufacturing**

Set & Sponsored by Dangote Cement Plc (DCP)

*Inaugural Edition*

Organised by the University of Lagos Engineering Society Academic & Research Board (ULES ARB)

## Participant Guidelines & FAQ

## 1. About the Challenge

The DCP University Engineering Challenge invites university engineering students to develop practical, evidence-based solutions to real operational problems faced in cement manufacturing. Working in teams, students choose one of four challenge tracks, diagnose the problem, and present an engineering and business case for a workable solution.

The challenge is set and sponsored by Dangote Cement Plc (DCP), whose operations include plants at Obajana, Ibese, Gboko and Okpella. It is designed to connect classroom learning with industry practice and to give students a structured way to apply engineering reasoning to genuine plant-level problems.

This challenge is designed to:

- Bridge the gap between academic engineering and real industrial operations
- Develop students' problem-solving, data, and business-case communication skills
- Surface fresh, practical ideas for cement manufacturing efficiency, reliability and sustainability
- Build links between DCP and emerging engineering talent across universities

## 2. Event Details

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| **Event Name** | DCP University Engineering Challenge |
| **Theme** | Engineering Solutions for Sustainable, Reliable and Efficient Cement Manufacturing |
| **Set & Sponsored By** | Dangote Cement Plc (DCP) |
| **Organised By** | University of Lagos Engineering Society Academic & Research Board (ULES ARB) |
| **Edition / Year** | Inaugural Edition |

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|---|---|
| **Format** | Team-based; one of four challenge tracks; written submission + pitch |
| **Registration Opens** | Monday, 29th June, 2026 |
| **Submission Deadline** | 11:59pm Thursday, 9th July, 2026 |
| **Shortlist Decision** | Sunday, 12th July, 2026 |
| **Final Pitch / Results** | Tuesday, 14th July, 2026 |
| **Contact** | arb.unilagengr@gmail.com \| +2348064627924 |

## 3. Eligibility

This challenge is open to currently enrolled university engineering students. To take part, teams should:

- Be made up of currently enrolled university students studying an engineering discipline (a multidisciplinary team is encouraged, as each track draws on more than one discipline)
- Be a team of **TWO - FOUR** members
- Select one of the four challenge tracks in Section 4
- Submit only their own original work; presenting another team's work is grounds for disqualification

*No prior cement-industry experience is required. The brief in Section 4 gives the context you need, and submissions are judged on the quality of your engineering reasoning, not on insider knowledge.*

## 4. The Four Challenge Tracks

Each team selects ONE of the following tracks. All four tracks are weighted equally; choose the one that best matches your team's strengths and disciplines.

### Challenge 1: Alternative Fuel Sourcing and Pricing Optimization

**Problem Statement**

DCP wants to improve alternative-fuel utilisation in its cement kilns, but sourcing, cost, consistency, quality, logistics and handling can be challenging. Teams are expected to identify feasible alternative-fuel streams within Nigeria and design a sourcing model that can deliver reliable, safe, and cost-competitive fuel to selected cement plants.

**Why this matters to DCP**

Alternative fuel directly supports the thermal substitution rate (TSR), fuel-cost optimisation, sustainability, circular-economy impact and $CO_2$ reduction. It is especially relevant where TSR is below target and where fuel availability or price competitiveness limits kiln substitution.

**Expected Deliverables**

- Alternative-fuel opportunity map by location and fuel type
- Cost-per-ton and basic energy-value comparison
- Recommended top three alternative-fuel options with justification
- Pilot implementation roadmap, risk register and mitigation plan
- Business case showing the expected value to DCP

**Recommended disciplines:** Chemical Engineering, Mechanical Engineering, Civil & Environmental Engineering, and others

## Challenge 2: Predictive Maintenance and Reliability Early-Warning System

**Problem Statement**

DCP plants run multiple critical equipment systems — crushers, raw mills, coal mills, kilns, cement mills and packing equipment. Unplanned failures and repeat stoppages reduce plant reliability, throughput and cost performance. Teams are expected to design a practical early-warning model that predicts or flags high-risk equipment before major failures occur.

**Why this matters to DCP**

Reliability is a recurring operational priority. A simple, practical predictive system helps maintenance and operations teams prioritise inspections, reduce repeat failures, improve root-cause-analysis discipline, and protect production output.

**Expected Deliverables**

- Reliability early-warning framework
- Risk-scoring model with clear assumptions
- Dashboard mock-up or prototype
- Example maintenance escalation workflow
- Implementation roadmap for plant use

**Recommended disciplines:** Mechanical Engineering, Electrical Engineering, Mechatronics, Computer Engineering, and others

## Challenge 3: Cement Production Conversion Gap

**Problem Statement**

A cement plant may have clinker available, or achieve reasonable clinker production, yet still miss cement-production targets. Teams are expected to diagnose why clinker strength does not always translate into cement output, and design a conversion-loss bridge that separates rate, runtime, feed, additive, packing, dispatch and planning constraints.

**Why this matters to DCP**

This addresses a real operating issue: cement production is not only a kiln problem. Output depends on mill availability, grinding rate, product mix, additive availability, silo space, packing capacity, dispatch readiness, power availability and production planning.

**Expected Deliverables**

- Clinker-to-cement conversion framework
- Loss tree or loss-bridge template
- Dashboard mock-up for daily plant review
- Root-cause hypotheses for typical conversion losses
- Action plan with owners by system

**Recommended disciplines:** Mechanical Engineering, Chemical Engineering, and others

## Challenge 4: Digital KPI Automation and Data Quality

### Problem Statement

DCP plant-performance reporting depends on timely and accurate daily, monthly and year-to-date KPI data. Manual reporting increases the risk of delays, inconsistent templates, missing numbers and weak traceability. Teams are expected to design a digital KPI reporting and validation concept that improves data quality, exception visibility and management decision-making.

### Why this matters to DCP

Accurate data is the foundation for plant-performance improvement. This track supports faster reporting, better governance, fewer manual errors, standard KPI definitions, and improved visibility across Obajana, Ibese, Gboko and Okpella.

### Expected Deliverables

- KPI data model and reporting workflow
- Validation rule library
- Exception dashboard / wireframe
- Plant sign-off and escalation process
- Implementation roadmap using Excel, Power BI, Power Apps, Python or other suitable tools

**Recommended disciplines:** Computer Engineering, Systems Engineering, Electrical/Electronic Engineering, and others

# 5. Final Submission Requirements

Each team is expected to submit the following for its chosen track:

- A 10-slide executive presentation
- A technical appendix or model/prototype, where applicable
- A one-page business case showing value, cost, assumptions and risks
- A simple implementation roadmap showing immediate, short-term and medium-term actions
- A risk and safety assessment relevant to the proposed solution
- A statement of assumptions and data limitations

### Recommended 10-Slide Pitch Format

The executive presentation should follow this slide-by-slide structure:

1. Challenge selected and why it matters to DCP
2. Current-state problem diagnosis
3. Data / exhibit insights and key assumptions
4. Root-cause hypothesis
5. Proposed engineering / business solution
6. Technical feasibility and operating concept
7. Business case and expected impact
8. Risk, safety, sustainability and mitigation plan
9. Implementation roadmap and resources required
10. Final recommendation and pilot proposal

## 6. Judging Criteria

Submissions will be evaluated by a panel of judges. The criteria below are aligned to the expected deliverables and the 10-slide pitch format; final weights will be confirmed by the organising team.

| Criterion | What judges look for | Weight |
|---|---|---|
| Problem Diagnosis & Insight | Depth of current-state analysis, quality of root-cause reasoning, and sound use of data and assumptions. | 30% |
| Solution Quality & Feasibility | Soundness of the proposed engineering/business solution and how workable it is in a real plant. | 30% |
| Business Case & Impact | Clear value to DCP - cost, expected impact, and realistic implementation roadmap. | 20% |
| Risk, Safety & Sustainability | Quality of the risk register, safety assessment and sustainability considerations. | 10% |
| Communication & Presentation | Clarity, structure and professionalism of the 10-slide pitch and supporting materials. | 10% |

*Judges' decisions are final. Scores and feedback may or may not be shared with participants, at the discretion of the organising committee.*

## 7. Prizes & Awards

Awards will be presented to the top teams at the conclusion of the challenge. Prize details will be confirmed by the organising team before the final pitch.

| Award | Prize | Recognition |
|---|---|---|
| 1st Place | 700,000 Naira | Certificate |
| 2nd Place | 500,000 Naira | Certificate |
| 3rd Place | 300,000 Naira | Certificate |

## 8. How to Register

Teams enter through the official registration portal: https://ulesarb.org/

Registration opens on **30th June, 2026** and closes on **9th July, 2026**. Late entries will not be accepted.

During registration, your team will indicate its chosen challenge track and team members. Submission instructions and templates will be shared with registered teams.

## 9. Frequently Asked Questions (FAQ)

**Q: Do I enter as an individual or as a team?**
A: This is a team-based challenge. Because each track draws on more than one engineering discipline, a multidisciplinary team is encouraged. The exact team size will be confirmed by the organising team.

**Q: Can students from different disciplines (or different departments) team up?**
A: Yes, it is encouraged. Each track lists recommended disciplines followed by "and others," so teams that combine complementary skills (for example, mechanical, electrical and computer engineering) are well placed.

**Q: Do I have to choose only one challenge, or can my team tackle several?**
A: Each team selects ONE of the four tracks. The pitch is built around a single challenge, so focus your effort on one and make the strongest possible case for it.

**Q: Which challenge should we pick? Are some worth more points than others?**
A: All four tracks are weighted equally. Choose the one that best fits your team's disciplines, interests and access to relevant data or tools, not one you think is "easier."

**Q: Do I need real DCP plant data to take part?**
A: No. You are not required to have confidential or internal DCP data. Use publicly available information, reasonable engineering assumptions, and clearly stated estimates. Two of the required deliverables, "key assumptions" and a "statement of assumptions and data limitations", exist precisely so you can show your reasoning where data is incomplete.

**Q: Do we have to build a working prototype or model?**
A: Not necessarily. The technical appendix or model/prototype is required "where applicable." A clear concept, framework, mock-up or wireframe can be enough — several tracks specifically ask for a dashboard mock-up rather than a finished system. Build a prototype only if it strengthens your case.

**Q: What tools are we allowed to use?**

A: Any suitable engineering or analysis tools. The brief explicitly mentions Excel, Power BI, Power Apps and Python for the KPI track, but you are free to use whatever helps you diagnose the problem and present a solid solution.

**Q: Do I need prior knowledge of cement manufacturing?**
A: No. Each track's problem statement and “why this matters” section gives you the context. You are judged on engineering reasoning, diagnosis and the quality of your proposed solution, not on insider industry knowledge.

**Q: What exactly do we submit at the end?**
A: Six items: a 10-slide executive presentation; a technical appendix or model/prototype (where applicable); a one-page business case; a simple implementation roadmap; a risk and safety assessment; and a statement of assumptions and data limitations. See Section 5 for the full list and the recommended 10-slide structure.

**Q: Is there a limit on the number of slides?**
A: Yes — the executive presentation is 10 slides, and Section 5 gives a recommended slide-by-slide structure. Supporting detail goes in the technical appendix, not in extra slides.

**Q: What are the four DCP plants mentioned, and do we have to focus on a specific one?**
A: Obajana, Ibese, Gboko and Okpella are DCP cement plants. Unless a track or the organisers state otherwise, you can frame your solution generally or focus on a representative plant — just state your scope and assumptions clearly.

**Q: The brief is marked “External Use.” Can we share our solution or discuss it publicly?**
A: “External Use” means the brief itself may be shared outside DCP, so you may discuss the challenge. Your submission should rely only on public information and your own assumptions — do not include any confidential or proprietary data, and follow your university's academic-integrity rules. Submitted work may be reviewed by DCP.

**Q: Do we present live, or only submit documents?**
A: The challenge includes an executive pitch, so plan to present your 10-slide deck. Whether the final pitch is live, virtual or recorded — and the exact format — will be confirmed by the organising team.

**Q: Is there a registration fee?**
A: Registration details, including any fee, will be confirmed by the organising team. Check the Event Details and How to Register sections for updates.

**Q: When are the deadlines?**
A: Key dates (registration, submission, shortlist and final pitch) are listed in Section 2 and will be confirmed before registration opens. Watch for the official announcement.

**Q: Who do we contact with more questions?**
A: Reach the organising team using the contact details in Section 2 and Section 11 once they are confirmed.

**Q: How many documents are to be submitted?**

A: Two documents. One Slide presentation for the executive presentation and another document encompassing the rest of the submission.

## 10. Tips for a Winning Submission

- Diagnose before you solve: judges reward a clear, data-grounded picture of the current-state problem and its root causes.
- Tie everything back to DCP value: show the cost, the impact, and why your solution matters operationally.
- State your assumptions openly: where data is missing, reasonable, clearly-labelled assumptions are a strength, not a weakness.
- Keep the deck to 10 focused slides: push supporting detail into the technical appendix.
- Make it implementable: a realistic, phased roadmap beats an ambitious idea with no path to delivery.
- Don't skip risk and safety: a credible risk register and safety assessment signals real engineering maturity.

## 11. Contact & Further Information

For all enquiries, please reach out to the organising team:

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| **Organising Body** | University of Lagos Engineering Society Academic & Research Board (ULES ARB) |
| **Email** | arb.unilagengr@gmail.com |
| **Phone / WhatsApp** | +2348064627924 |

## 12. About the Sponsor

This challenge is set and sponsored by Dangote Cement Plc (DCP). DCP is Africa's largest cement manufacturer with presence in 11 African countries and 4 integrated plants in Nigeria in Obajana, Ibese, Gboko and Okpella. The four challenge tracks reflect real operational priorities across DCP's plants.

*Good luck to all participating teams — we look forward to seeing your engineering solutions for sustainable, reliable and efficient cement manufacturing.*